Equations for modeling of the dynamics of fluids

Lecture for the course HL2008, Spring 2023

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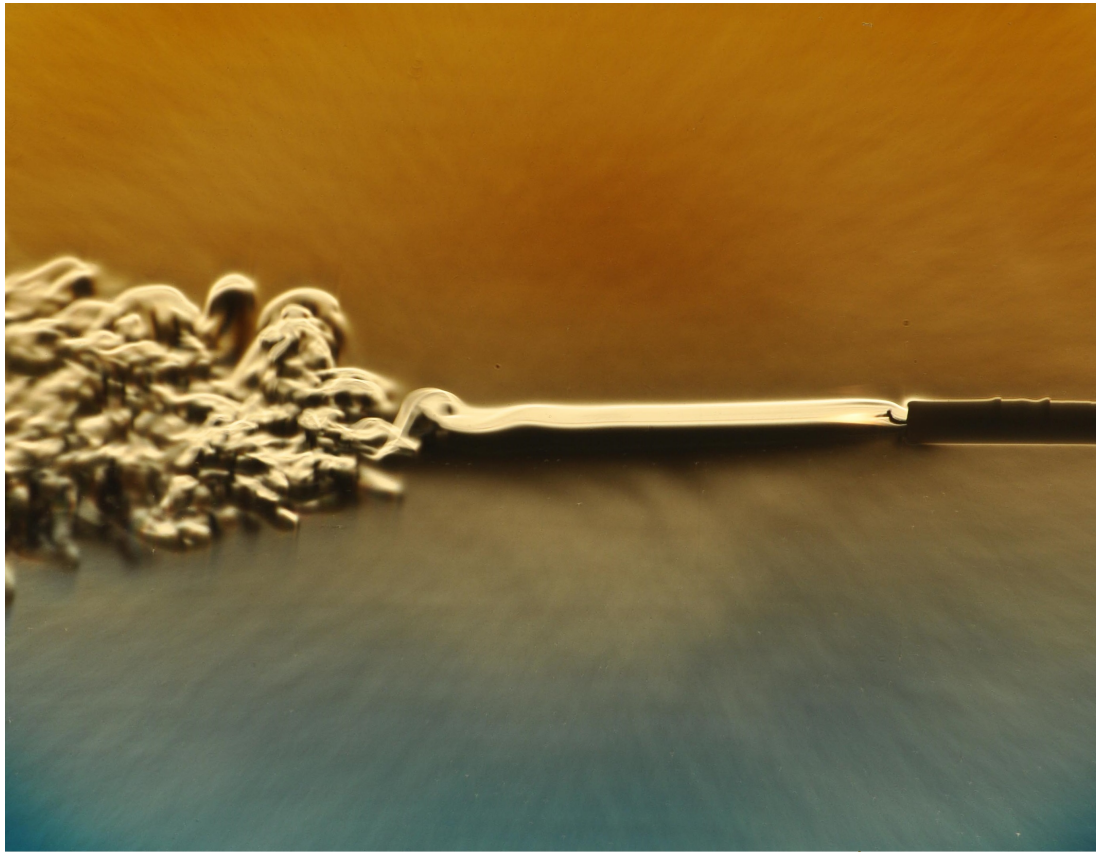
13th of February, 2023

Plan

- **Lecture today**: vector calculus, Navier-Stokes equations and some related concepts.
- **Laboratory assignment**: simulations in Maple (some animations) and some "pen and paper" work with simple case of Navier-Stokes equations. Each group will present the solutions in a short meeting which you sign up for.
- **Laboratory assignment "pass with exellence"**: you solve an extra task and present it during the meeting. Produce a derivation of one of the important equations in the lecture.

Fluid flows

We will talk about equations modelling fluid flows.



Assumptions

We will study fluid flows that satisfy some simplifying assumptions. They are

1. **incompressible**, in the sense that the fluid density ρ does not change with pressure,
2. **isothermal**, in the sense that the temperature of the fluid does not change,
3. have constant **viscosity** μ.

In particular, we will work with so called **Newtonian fluids** (our fluids are Newtonian by the assumption that the viscosity μ is constant).

Water and peanut butter

There are no perfectly Newtonian fluids in reality. In normal real life circumstances, **water** and **air** are Newtonian. A *non-Newtonian* "fluid" is peanut butter!

Newtonian



Non-Newtonian



A little review of notation

A function $\vec{V} : \mathbb{R}^3 \to \mathbb{R}^3$ is a **vector field** and it is described by three components:

$$\vec{V}(x, y, z) = \Big(u(x, y, z), v(x, y, z), w(x, y, z)\Big).$$

Our vector fields will often be **time-dependent as well**, so strictly speaking then they would be mappings $\mathbb{R}^4 \to \mathbb{R}^3$:

$$\vec{V}(x, y, z, t) = \Big(u(x, y, z, t), v(x, y, z, t), w(x, y, z, t)\Big).$$

The **velocity of the fluids** we are modelling will be time-dependent vector fields in $\mathbb{R}^3$. Show Maple animation

A little review of notation

A function $P : \mathbb{R}^3 \to \mathbb{R}$ is a **scalar field**. Scalar-fields can also be time-dependent, then strictly speaking they are functions $\mathbb{R}^4 \to \mathbb{R}$:

$$(x, y, z, t) \mapsto P(x, y, z, t).$$

The **pressure** P at points in the fluid we are modelling will be a time-dependent scalar field.

A little review of notation

- The **gradient** $\nabla\phi$ of a scalar field $\phi(x, y, z)$ is the vector field

$$\nabla\phi(x, y, z) = \left(\frac{\partial\phi}{\partial x}(x, y, z), \frac{\partial\phi}{\partial y}(x, y, z), \frac{\partial\phi}{\partial z}(x, y, z)\right).$$

- The **divergence** $\nabla \cdot \vec{V}$ of a vector field $\vec{V} = (u, v, w)$ is the scalar field

$$\nabla \cdot \vec{V} = \frac{\partial u}{\partial x}(x, y, z) + \frac{\partial v}{\partial y}(x, y, z) + \frac{\partial w}{\partial z}(x, y, z).$$

- The **Laplacian** $\Delta\phi$ of a scalar field $\phi(x, y, z,)$ is the scalar field

$$\Delta\phi(x, y, z) = \frac{\partial^2\phi}{\partial x^2}(x, y, z) + \frac{\partial^2\phi}{\partial y^2}(x, y, z) + \frac{\partial^2\phi}{\partial z^2}(x, y, z).$$

Continuity equation

The divergence $\nabla \cdot \vec{V}$ has an important physical interpretation. For a fluid of constant density with velocity field $\vec{V}$, the **conservation of mass principle** reduces to

$$\nabla \cdot \vec{V} = 0.$$

(Derivation on blackboard?) This is also known as the **continuity equation (for incompressible fluids)**, and there is a more general one for compressible fluids.

The Navier-Stokes equations

Navier-Stokes equations

$$\begin{cases} \nabla \cdot \vec{V} = 0 & \text{(cont. eq)} \\ \rho \frac{D\vec{V}}{Dt} = -\nabla P + \rho \vec{g} + \mu \Delta \vec{V} & \text{(N-S eq)} \end{cases}$$

Here

- P and $\vec{V}$ are the pressure and velocity fields of the fluid,
- ρ and μ are the constant density and viscosity,
- and $\vec{g}$ is the force of gravity.

The Navier-Stokes equations

Note that

- the second equation (N-S eq) is actually a vector equation, so three scalar equations,
- Laplacian $\Delta \vec{V} = (\Delta u, \Delta v, \Delta w)$ is applied to each component,
- $\frac{D\vec{V}}{Dt}$ is the **material derivative** defined as

$$\left(\frac{\partial}{\partial t} + \vec{V} \cdot \nabla\right) \vec{V}$$

and is also applied to each component.

A little on the material derivative

Consider the position vector

$$\vec{x}(t) = \Big(x(t), y(t), z(t)\Big)$$

of a little particle of the fluid that travels under the law imposed by the flow $\vec{V} = (u, v, w)$. The particle will follow its **pathline** Show Maple animation. The time derivative of the position vector is the velocity vector of the particle:

$$\frac{d\vec{x}}{dt}(t) = \Big(\frac{dx}{dt}(t), \frac{dy}{dt}(t), \frac{dz}{dt}(t)\Big)$$

which by assumption equals

$$\vec{V}(\vec{x}(t), t) = \Big(u(\vec{x}(t), t), v(\vec{x}(t), t), w(\vec{x}(t), t)\Big).$$

A little on the material derivative

Differentiating again leads to the acceleration

$$\frac{d^2\vec{x}}{dt^2}(t)$$

of the particle. The first component of $\frac{d^2\vec{x}}{dt^2}(t)$ equals

$$\frac{d}{dt}u(\vec{x}(t),t) = \frac{\partial u}{\partial t} + \frac{d\vec{x}}{dt}\cdot\nabla u = \frac{\partial u}{\partial t} + \vec{V}\cdot\nabla u,$$

which is the first component of the material derivative:

$$\frac{D\vec{V}}{Dt} = \left(\frac{\partial}{\partial t} + \vec{V}\cdot\nabla\right)\vec{V}.$$

Cauchy momentum equation

Essentially, the Navier-Stokes equations follow from the

Cauchy momentum equation

$$\rho \frac{D\vec{V}}{Dt} = \nabla \cdot \boldsymbol{\sigma} + \rho \vec{g}$$

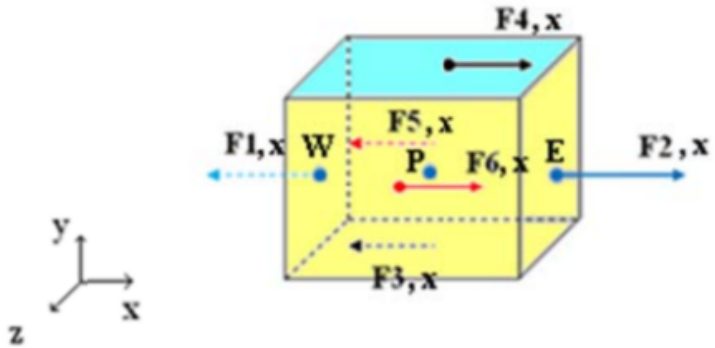
applied to Newtonian fluids. The equation describes the motion of a small volume of fluid. Here $\vec{g}$ denotes the force of gravity and $\boldsymbol{\sigma}$ is the **Cauchy stress tensor**

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}$$

which is a symmetric matrix, different at each point (x, y, z), and describes the acting **surface forces** (pressure and viscous forces).

Derivation of the Cauchy equations

Cauchy's equation in turn follows from **Newton's second law of motion**. We will sketch a derivation. Let us envision a small box-shaped volume of the fluid with dimensions dx, dy, dz:



According to Newton's second law, we have

$$m\vec{a} = \rho dxdydz\frac{D\vec{V}}{Dt} = \vec{F}.$$

Derivation of the Cauchy equations

We can write

$$\vec{F} = \rho dxdydz\vec{g} + \vec{F_s},$$

where $\vec{F_s}$ are the **surface forces** (pressure and viscous forces), which are described with the Cauchy stress tensor $\boldsymbol{\sigma}$. The surface force on the face of the volume with normal vector $\vec{n}$ and area A equals

$$A\boldsymbol{\sigma}\vec{n}.$$

Applying this to each of the six faces of the box, together with some differential calculus (blackboard derivation?) volume gives the Cauchy momentum equation.

Derivation of the N-S equations

To get to the Navier-Stokes equation from Cauchy equations we will decompose the stress tensor $\boldsymbol{\sigma}$ as

$$\boldsymbol{\sigma} = \boldsymbol{\tau} + \boldsymbol{p},$$

where $\boldsymbol{p} = -P \cdot I$ is the **pressure tensor** which is a diagonal matrix with elements $-P$ on the diagonal, and $\boldsymbol{\tau}$ is the **viscous stress tensor**

$$\boldsymbol{\tau} = \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix}$$

Derivation of the N-S equations

Now we use another law from physics.

Newton's law of viscosity (for incompressible fluids of constant viscosity)

If $\vec{V} = (u, v, w) = (V_x, V_y, V_z)$ is the flow velocity, then

$$\tau_{xx} = \mu\Big(\frac{\partial V_x}{\partial x} + \frac{\partial V_x}{\partial x}\Big),$$

$$\tau_{xy} = \mu\Big(\frac{\partial V_x}{\partial y} + \frac{\partial V_y}{\partial x}\Big),$$

$$\tau_{xz} = \mu\Big(\frac{\partial V_x}{\partial z} + \frac{\partial V_z}{\partial x}\Big)$$

and similar for the other 6 components of the matrix $\boldsymbol{\tau}$.

Derivation of the N-S equations

Note that, using the viscosity law, a simple computation (blackboard derivation?) gives

$$\nabla \cdot \left(\tau_{xx}, \tau_{xy}, \tau_{xz}\right) = \mu \Delta V_x + \frac{\partial}{\partial x}\left(\nabla \cdot \vec{V}\right) = \mu \Delta V_x,$$

and so we have

$$\vec{F_s} = \nabla \cdot (\boldsymbol{\tau} + \boldsymbol{p}) = \mu \Delta \vec{V} + \nabla P.$$

Putting this into the Cauchy equation gives the Navier-Stokes equations.

Thank you!